

Concept Paper: Secure AI Adoption & PDPA Compliance for New Telecom Ltd.

1. Deconstruction of the Problem

New Telecom Ltd. is at a critical juncture, transitioning from a traditional telecommunications operator to a digital service provider powered by Artificial Intelligence. While the organization possesses a strong baseline of cybersecurity controls, its privacy governance remains fragmented across business units. The introduction of the Bangladesh Personal Data Protection Act (PDPA) and the rapid deployment of AI (e.g., chatbots, predictive models) create a complex challenge: the company must innovate and extract value from vast data lakes without compromising customer trust, violating strict new legal mandates, or losing visibility over intricate, cross-border data flows.

2. Key Privacy Challenges

- **Visibility & Fragmentation:** Data lineage is obscured across internal units and third-party AI vendors, making it impossible to answer where data lives or who processes it.
- **AI Security & Ethical Risks:** AI models introduce vulnerabilities like prompt injection and model inversion. Furthermore, "black box" automated decisions (e.g., churn prediction, fraud) risk algorithmic bias and lack the explainability required by regulators.
- **Strict PDPA Compliance:** The organization faces stringent mandates for consent management, data minimization, and a zero-tolerance prohibition on tracking or profiling minors (under 18).

3. Proposed Approach

We propose the **"Trust-First AI & Privacy Hub" (TAPH)**, a framework that moves beyond reactive compliance to proactive, customer-centric privacy engineering.

- **Automated Data Lineage & Unified CMP:** Deploy AI-driven Data Security Posture Management (DSPM) to dynamically map data flows from customer to vendor. Couple this with a centralized Consent Management Platform (CMP) on the telecom's app, giving customers granular, transparent control over their data.
- **Privacy-Enhancing Technologies (PETs):** Mandate the use of Differential Privacy and Synthetic Data for training AI models (e.g., marketing engines, network optimization). This ensures raw Personally Identifiable Information (PII) is never exposed to AI training environments or third-party vendors.
- **Explainable AI (XAI) & Impact Assessments:** Require XAI frameworks for all automated decision-making to ensure human-readable transparency. Institute mandatory Algorithmic Impact Assessments (AIA) before any AI deployment to eliminate bias.
- **NID-Integrated "Zero-Tracking" Architecture:** To strictly comply with the PDPA's child protection mandate, integrate Bangladesh National ID (NID) verification at the subscriber level. Users under 18 will be automatically routed into a hardcoded "Zero-Tracking" group, technically blocking their data from marketing and behavioral analytics.
- **Federated Governance:** Establish a Hub-and-Spoke model led by a Chief Privacy Officer, with embedded "Privacy Champions" in AI, marketing, and IT teams to ensure localized accountability.

4. Expected Impact

- **Customer Trust as a Differentiator:** By providing a transparent CMP and explainable AI, privacy transforms from a legal footnote into a brand asset, empowering users and deepening brand loyalty.
- **Organizational Resilience:** PETs and automated mapping drastically reduce the "blast radius" of potential cyber breaches and ensure immediate audit-readiness for PDPA regulators.
- **Responsible Innovation:** Developers can innovate rapidly using synthetic data without waiting for complex compliance approvals, accelerating time-to-market for digital services.

5. Applicability

In Bangladesh's rapidly digitizing economy, this framework sets a gold standard for PDPA compliance. By leveraging the local NID system for age verification, it provides a uniquely localized, practical solution to protect minors. Furthermore, keeping sensitive telecom data within sovereign borders while utilizing anonymized data for global AI tools aligns with national security interests, fostering a safe, ethical digital ecosystem for millions of Bangladeshi citizens.